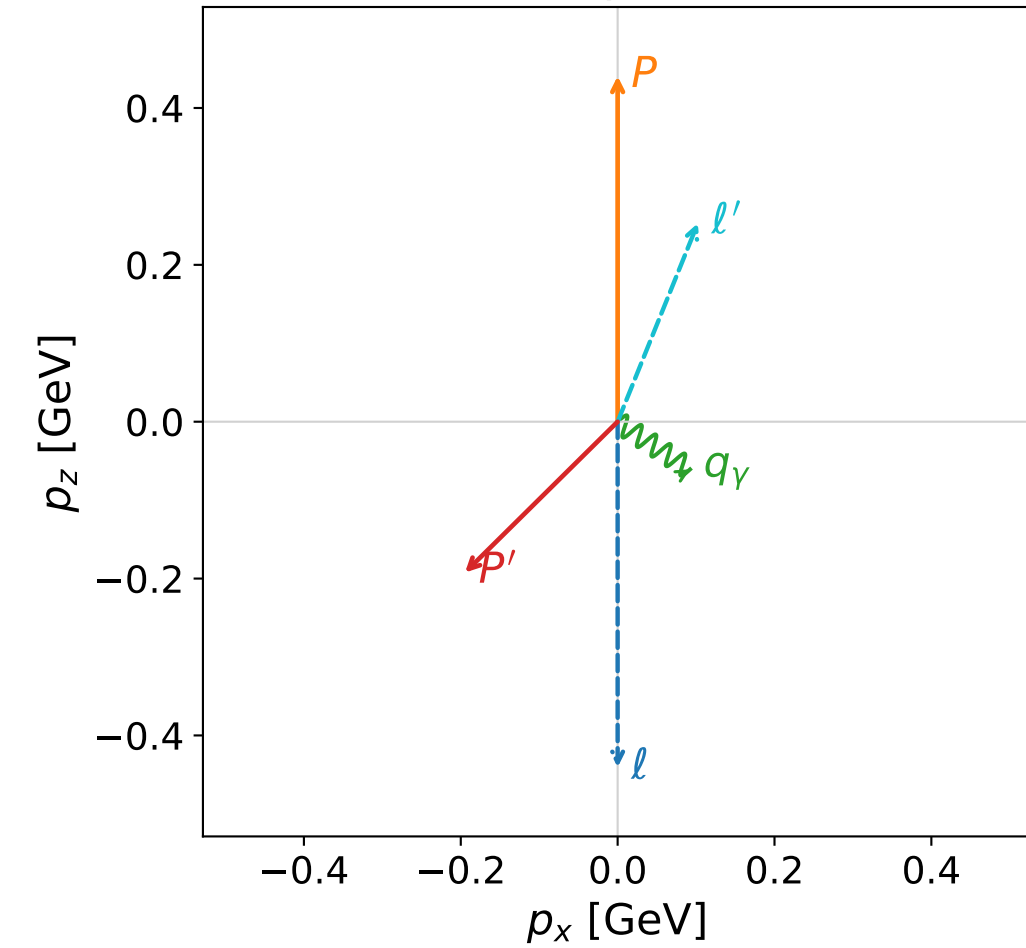
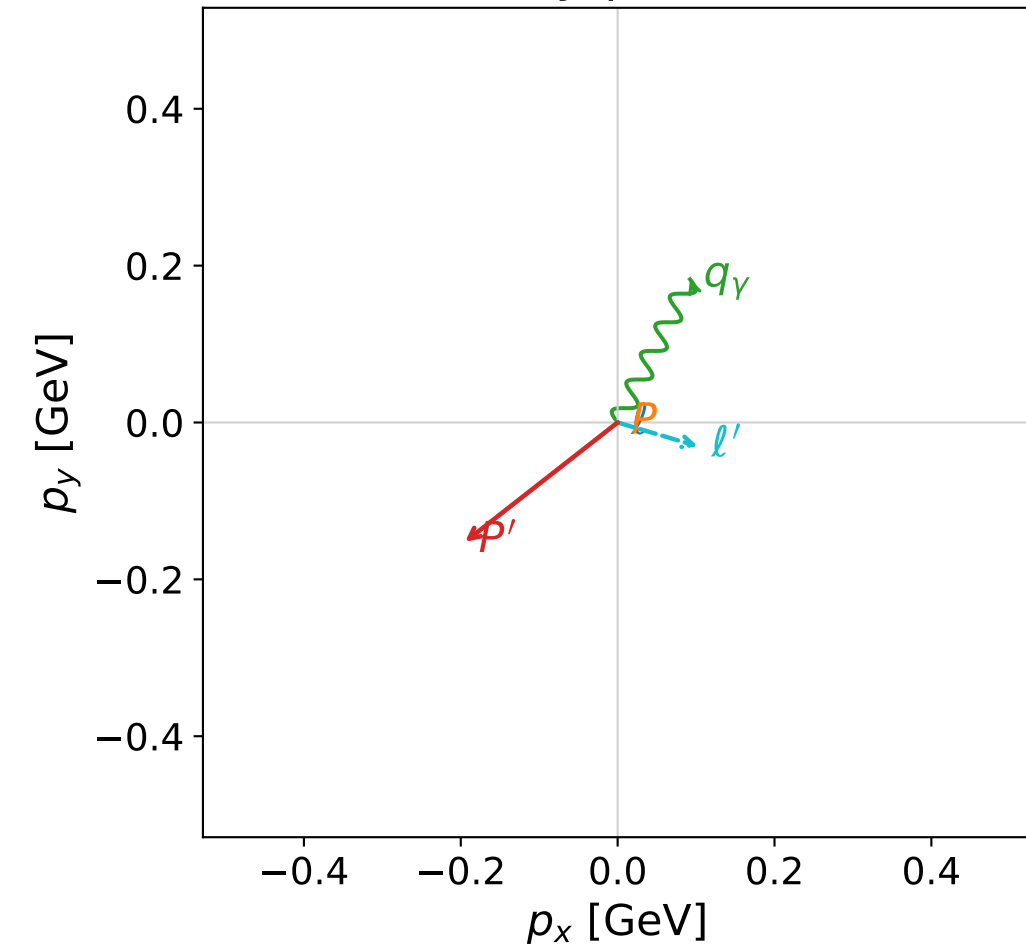


coherent mixing angles: minimum D_W configuration (gradient_local_minimum)

z--x plane



x--y plane



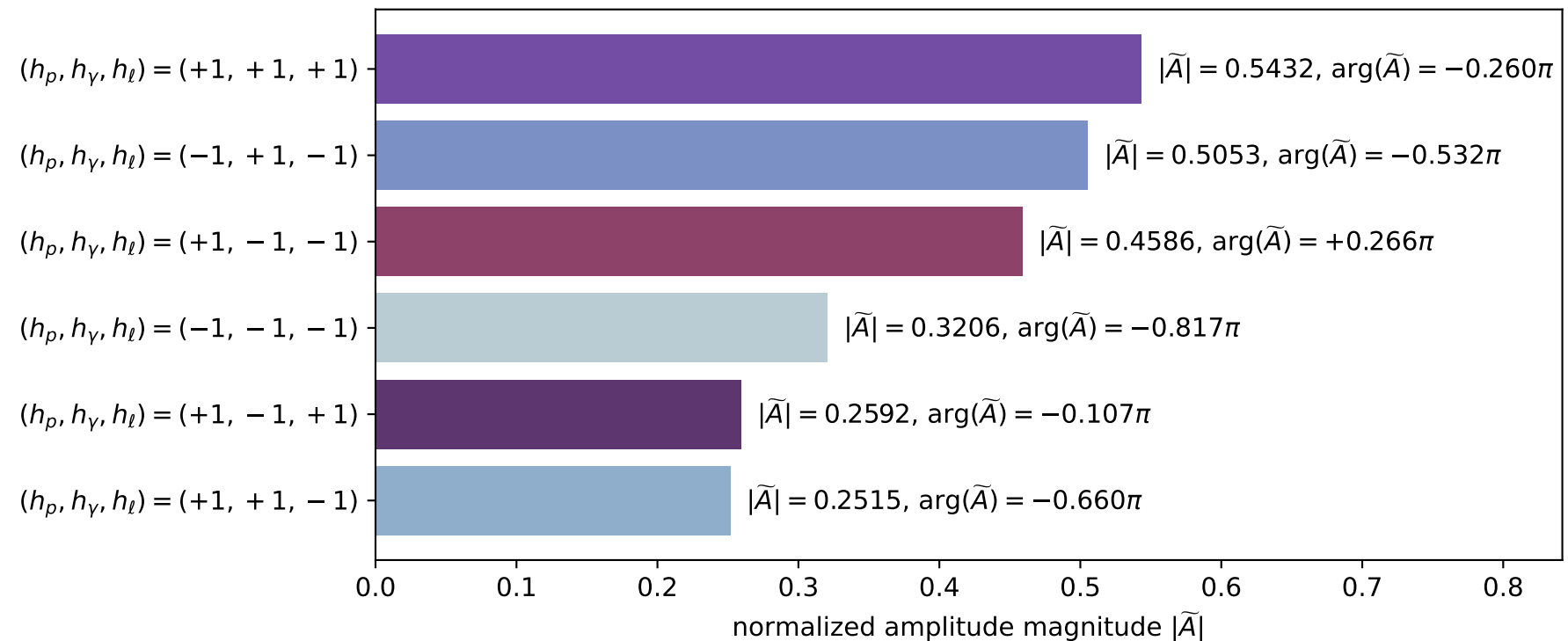
dw_mixing_angles_polarization_04_local_minimum_1
 $D_W=2.27876e-05$
 region: polarization_cluster_04_local_minimum_1
 outgoing-state purity: 1
 kinematic point: gradient_local_minimum
 lepton mixing angle: $\alpha_e=1.4456912$ rad
 incoming lepton: $0.124779|+\rangle +0.992185|-\rangle$
 proton mixing angle: $\alpha_p=3.0471872$ rad
 incoming proton: $-0.995547|+\rangle +0.0942653|-\rangle$

$s=2.18146$, $\sqrt{s}=1.47698$, $|\vec{P}|=0.440635$, $|\vec{P}'|=0.313109$
 $\theta_{p'}=2.23228$, $\theta_\gamma=1.85972$, $\phi_{p'}=3.80543$, $\phi_\gamma=1.10266$
 $E_\gamma=0.213391$, $\theta_{l'}=0.398906$, $\phi_{l'}=5.99517$
 $Q^2=0.465173$, $x_B=0.48693$, $t=-0.459446$, $W^2=1.36999$, $y=0.733948$

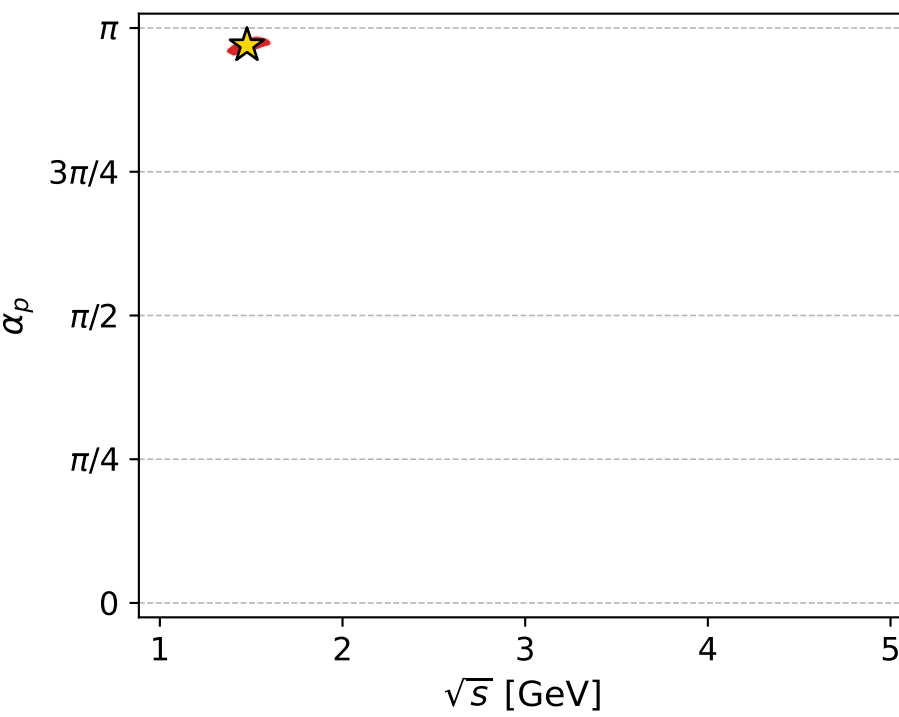
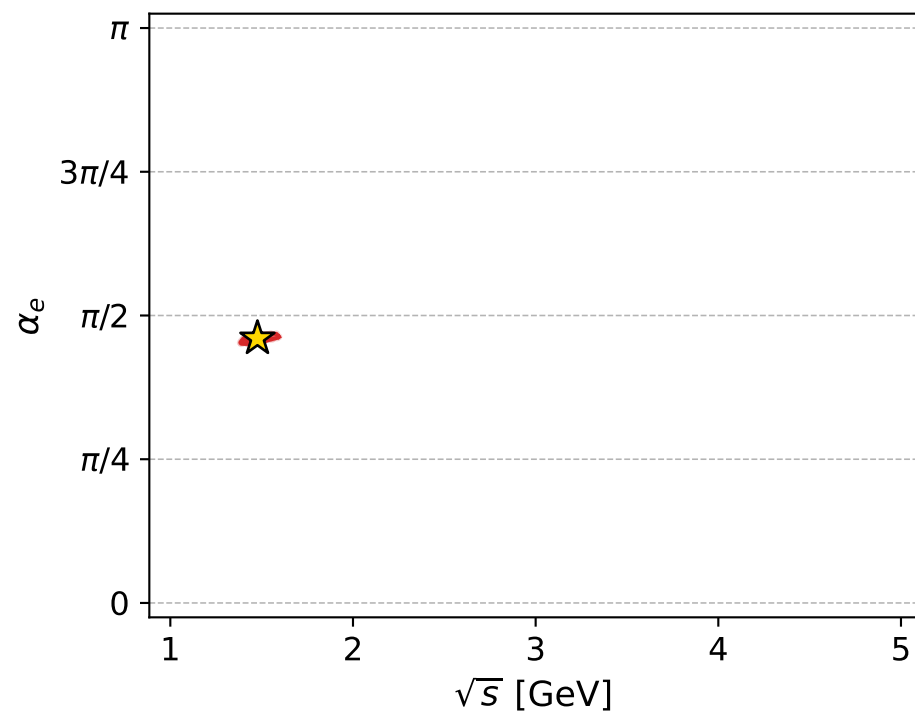
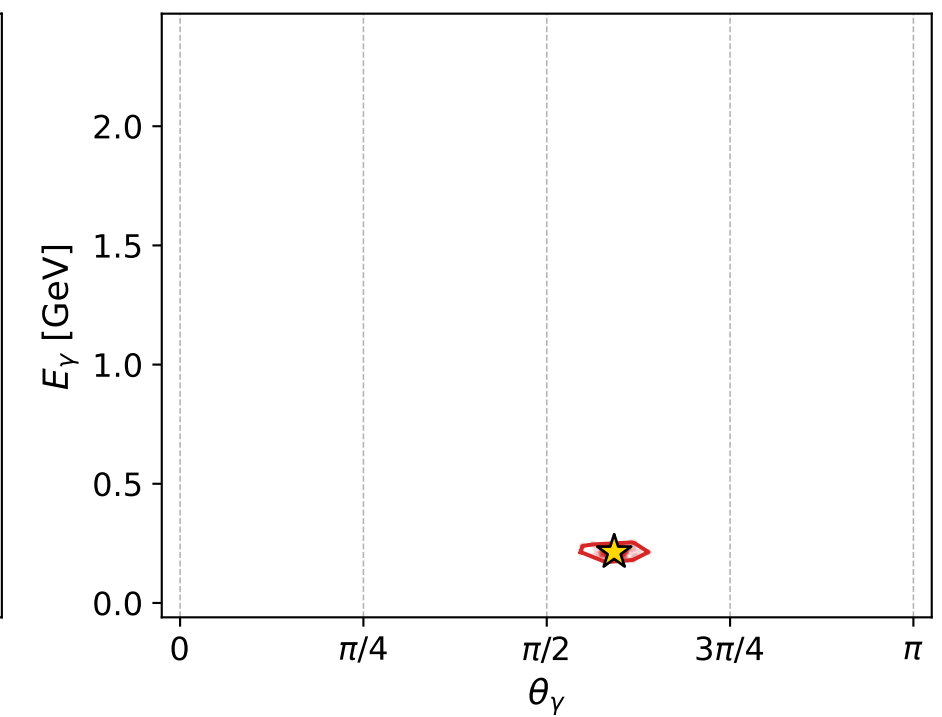
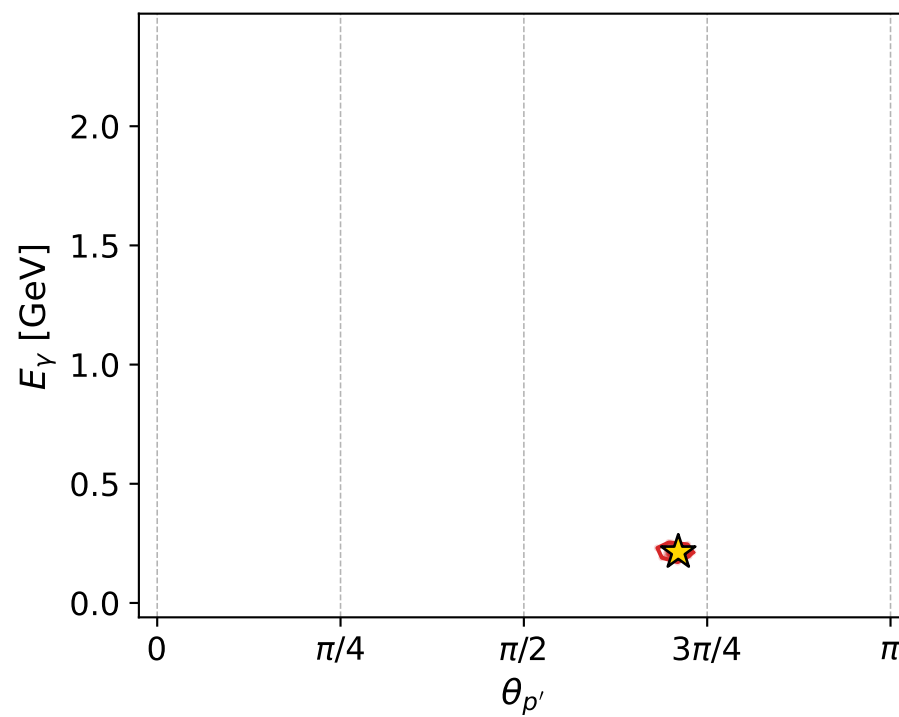
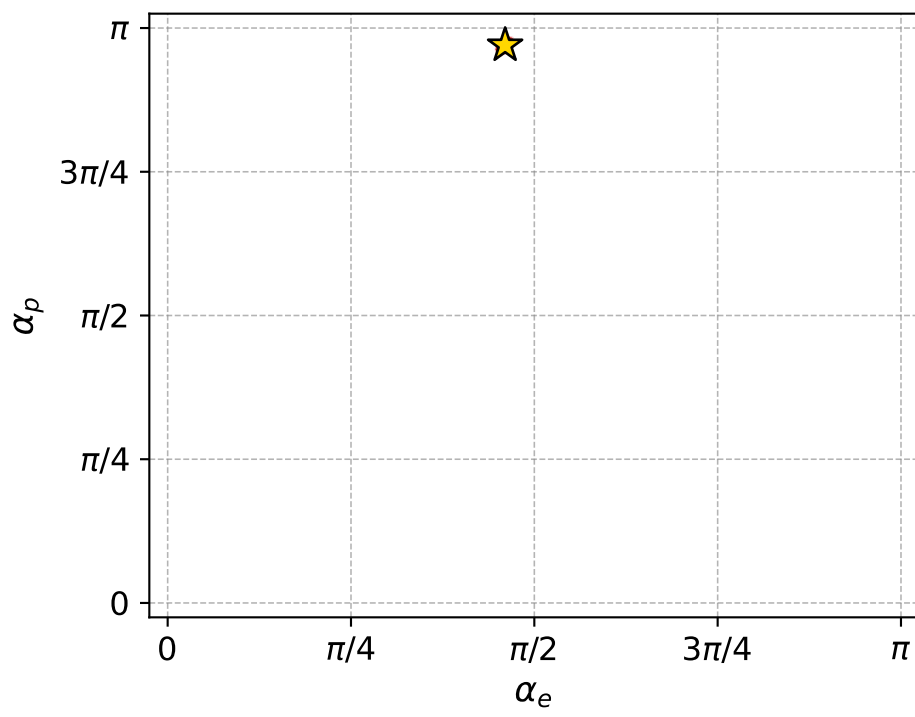
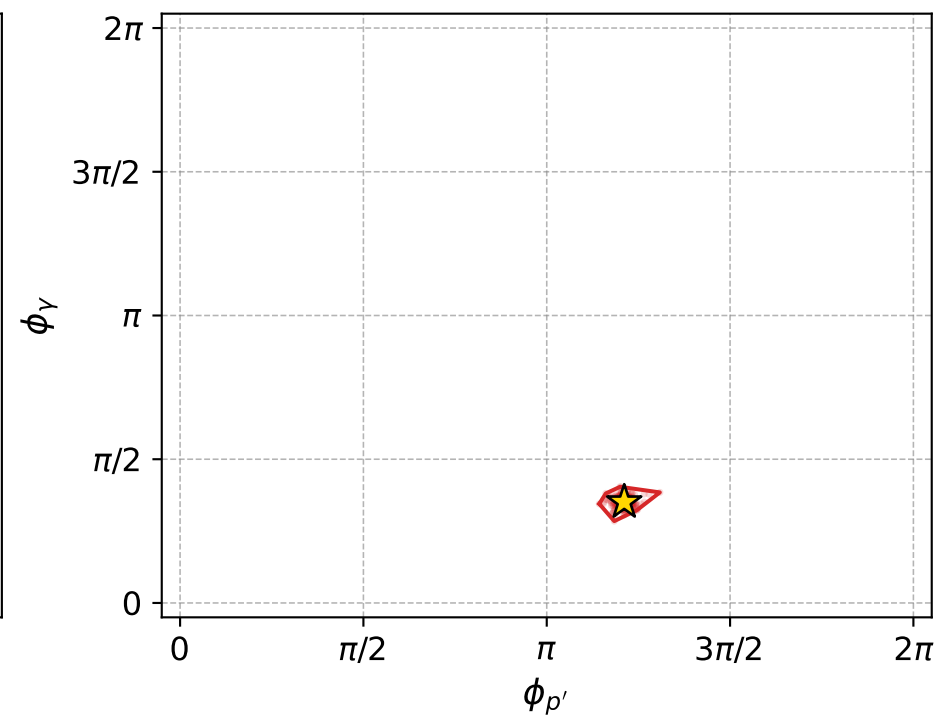
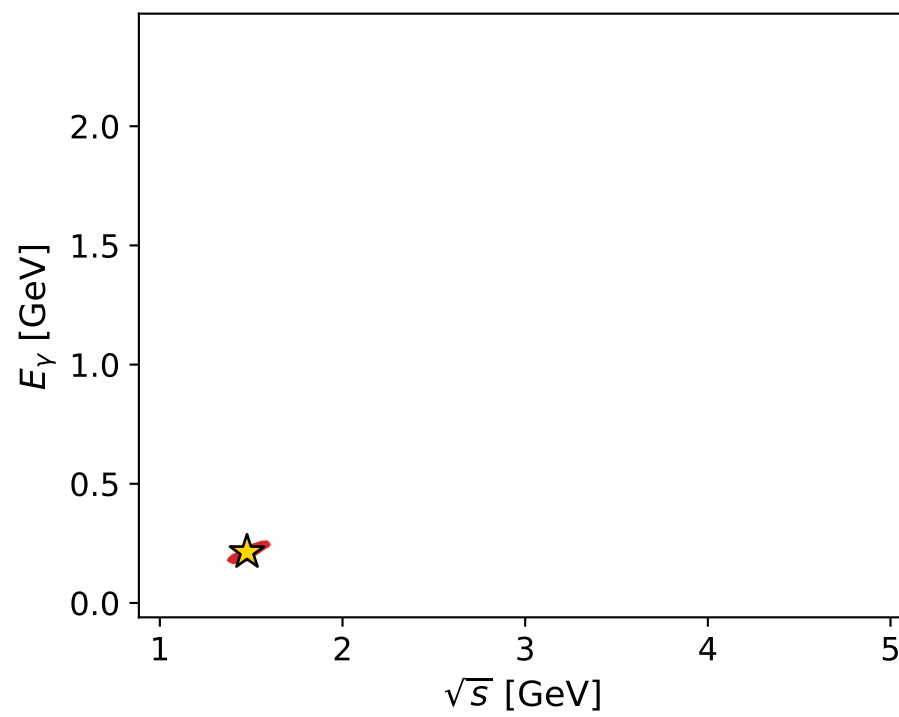
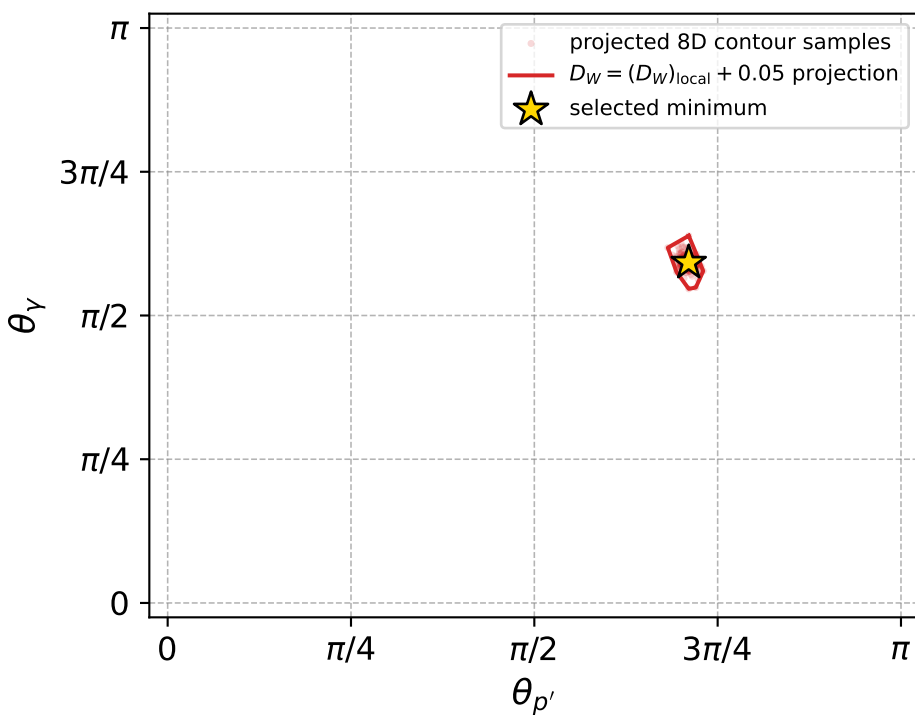
four-momenta $[E, p_x, p_y, p_z]$ GeV:

l	$E=$	0.4406	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	-0.4406
P	$E=$	1.0363	$p_x=$	0.0000	$p_y=$	0.0000	$p_z=$	0.4406
l'	$E=$	0.2747	$p_x=$	0.1023	$p_y=$	-0.0303	$p_z=$	0.2531
P'	$E=$	0.9889	$p_x=$	-0.1946	$p_y=$	-0.1522	$p_z=$	-0.1923
q_γ	$E=$	0.2134	$p_x=$	0.0923	$p_y=$	0.1825	$p_z=$	-0.0608

Leading normalized final-state amplitudes
 (N=6, retained=99.4%)



electron: polarization cluster P4, local minimum 1; pairwise projections of the 8D $D_W = (D_W)_{\text{local}} + 0.05$ contour



selected local minimum

$D_W(\text{local}) = 2.2787598\text{e-}05$
 D_W contour = 0.050022788
 above global minimum = 6.3547583e-06
 8D contour samples = 1536
 configured 8D directions = 1536

selected phase-space point:

$\text{sqrt_s} = 1.476976$
 $\text{theta_p_out} = 2.232277$
 $\text{theta_gamma_out} = 1.85972$
 $\text{q0out} = 0.2133906$
 $\text{phi_p_out} = 3.80543$
 $\text{phi_gamma_out} = 1.102656$
 $\alpha_e = 1.445691$
 $\alpha_p = 3.047187$